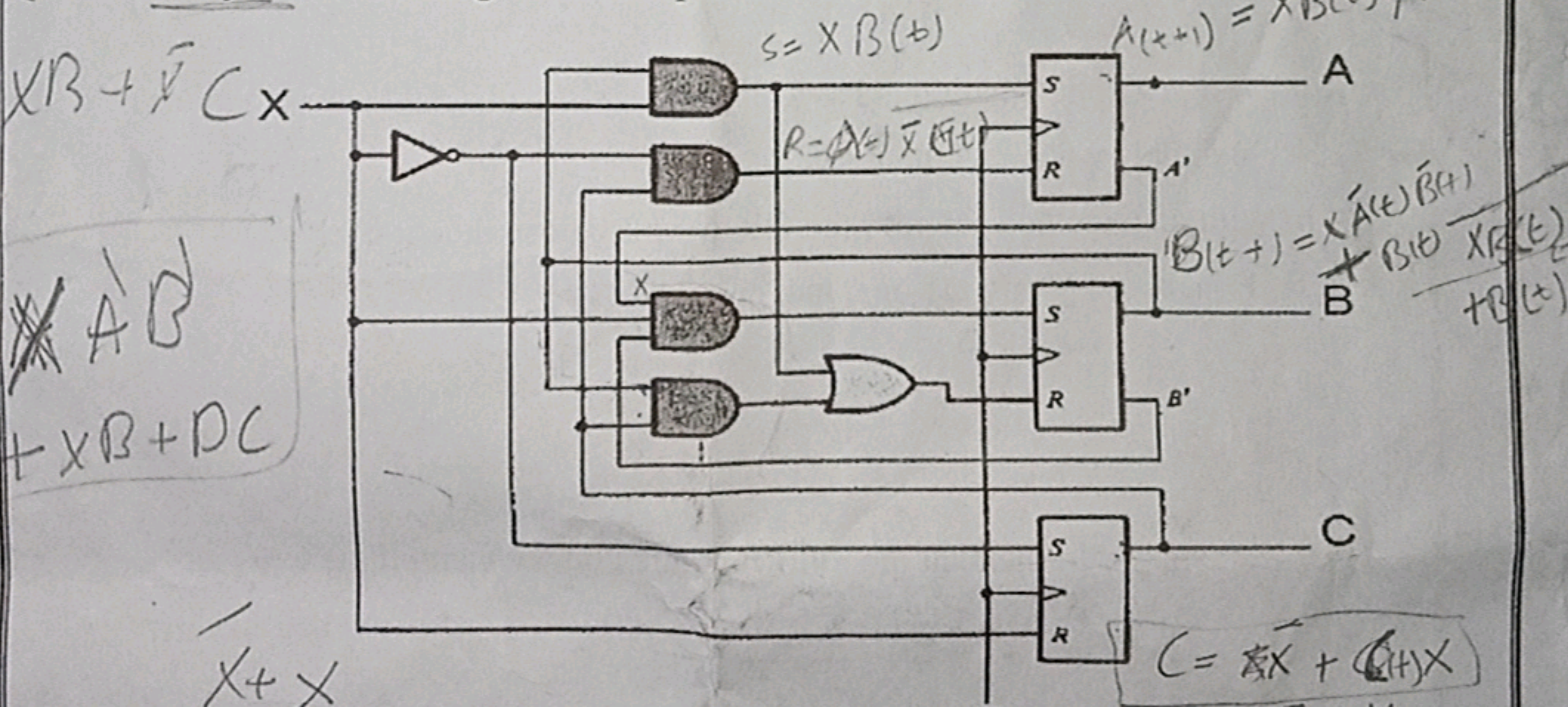


Q5

True or False:

- 1) A NOT gate have more than one input. $\times$
- 2) If any input to an OR gate is zero, the output is zero. $\times$
- 3) If all inputs to an AND gate are 1, the output is 1. $\checkmark$
- 4) A NAND gate can be considered as an AND gate followed by a NOT gate. $\checkmark$
- 5) A NOR gate can be considered as an OR gate followed by an inverter. $\checkmark$
- 6) The output of an exclusive-OR is 0 if the inputs are same. $\checkmark$ ^{XOR}
- 7) Addition in Boolean algebra is equivalent to the OR function. $\checkmark$
- 8) A multiplexer is a logic circuit that allows digital information from a single source to be routed onto several lines. $\times$
- 9) A half-adder adds two binary bits $\checkmark$
- 10) When the input bits are both 1, the sum output of a Half Adder is 1. $\times$

Q6 Analyze the following clocked sequential circuit. Obtain A; B; C



Note: - Question No. **1** is Compulsory.

- and answer any three questions of the rest

$$A'B'CD' + A'B'C'D + A'BCD' + AB'CD' + ABCD' + ABCD$$

a. Consider the following Boolean function, and answer the following questions:

$$F = (m_0 + m_1 + m_6 + m_{10} + m_{12} + m_{14})$$

- Construct the truth table corresponding to this logical function.
- Draw the direct (sum-of-product) circuit corresponding to this Function

b. Draw a circuit diagram for the (clocked) RS flip-flop and its truth table.

a. Design a logic circuit that has 4- inputs, x, y, z and d, and whose output (F) will be high only when all input same?

b. With map method simplify the following Boolean function in:

$$F(a, b, c) = \sum m_0; m_1, m_3, m_4, m_5$$

Implement the following function with a multiplexer:

$$F(a, b, c, d) = \sum (1, 2, 3, 4; 6; 7, 9, 11, 15) \text{ and draw the circuit?}$$